University of Cape Town



Machine Learning Assignment 1

Earthquake Analysis

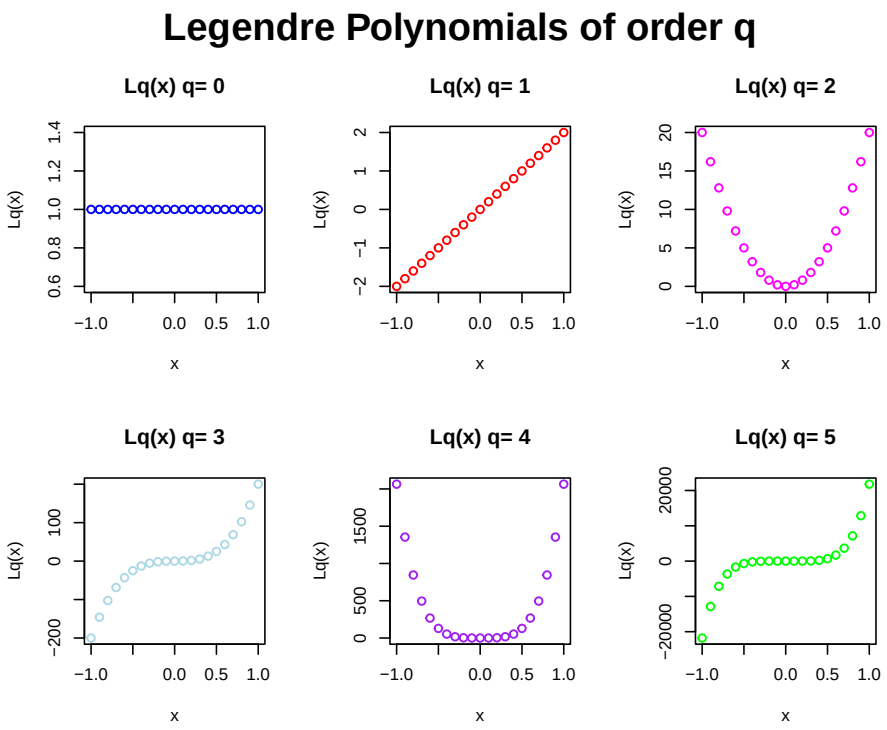
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Question 1

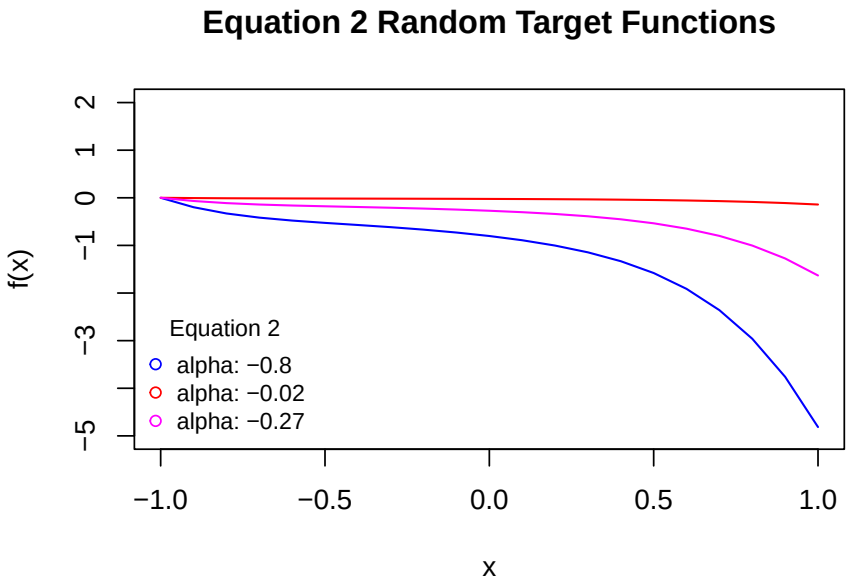
a) Legendre Polynomial Plots

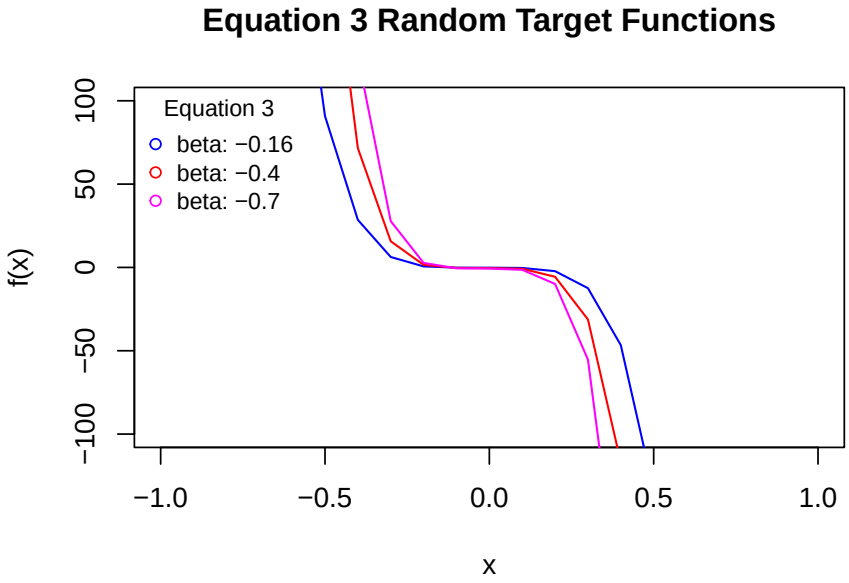


As the order of the polynomial, q, increases, the more nonlinear/complex the$L_q(x)$ gets, albeit slowly.

1. At q=0, $L_q(x)$ devolves to a straight horizontal line $L_q(x)$ =0 over the domain [-1,1].
2. At q=1 $L_q(x)$ starts to have some slope attached, a straight line of the form $y = 2x$
3. At q=2, 4 the non-linear curvature/complexity of $L_q(x)$ starts to become more visible particularly towards the edges of the domain [-1,1; function approximates a parabola/quadratic function
4. At q=3,5 non-linear curvature/complexity of $L_q(x)$ both take the same shape, albeit with different magnitudes of $L_q(x)$ over the same domain.

b) Random Target Functions



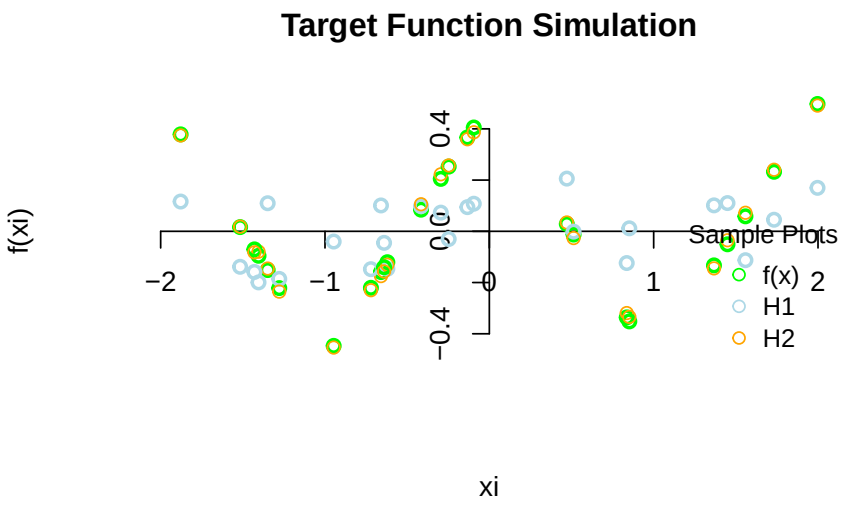


Observations:

1. For $f(x) = \sum_{q=0}^{Q_f} \alpha_q x^q$ the functions, looks to only ouput negative values over the domian [-1,1]
2. $f(x) = \sum_{q=0}^{Q_f} \beta_q L_q(x)$ outputs both positive and negative values *come-back

Question 3

a) Hypothesis Function Simulations

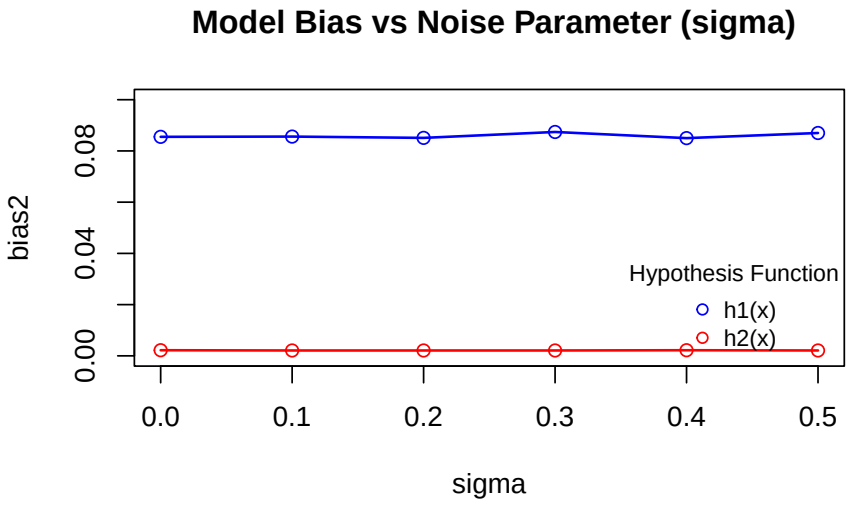


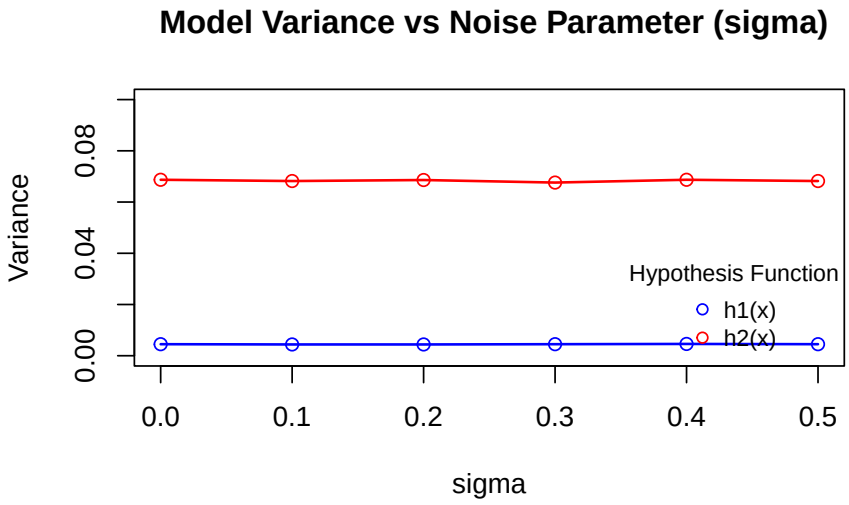
b) A priori expectation

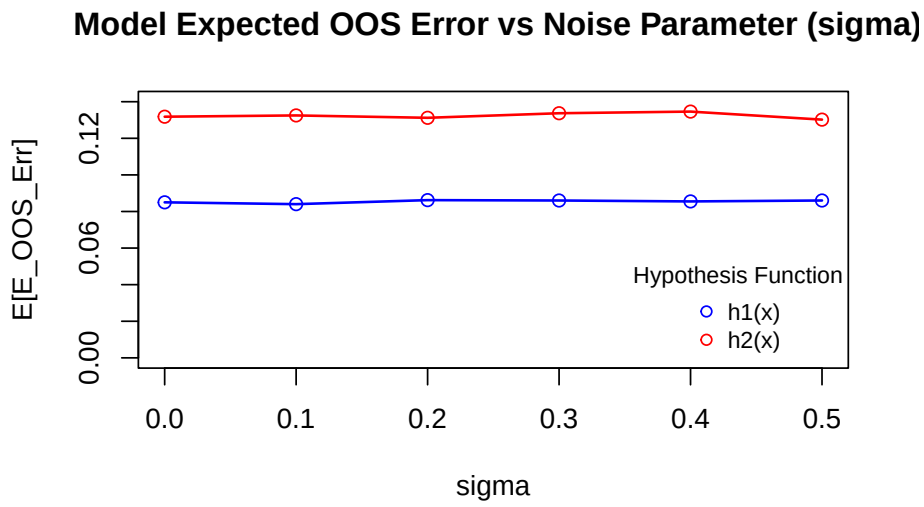
Given that the underlying data generating process is non-linear, the non-linear hypothesis would be more suited to fit the data's non-linear behavior or overall shape

c) Bias Variance for H1 and H2

	Bias	Variance	Both:Bias+Var	OOS Error	Estimate Differance
h1(x)	0.0854	0.0044	0.0898	0.0841	0.0057
h2(x)	0.0022	0.0703	0.0725	0.1296	0.0571


Model Bias vs Noise Parameter (sigma)
bias2
0.00
0.04
0.08
0.0
0.1
0.2
0.3
0.4
0.5
sigma
Hypothesis Function
h1(x)
h2(x)


Model Variance vs Noise Parameter (sigma)
Variance
0.00
0.04
0.08
0.0
0.1
0.2
0.3
0.4
0.5
sigma
Hypothesis Function
h1(x)
h2(x)

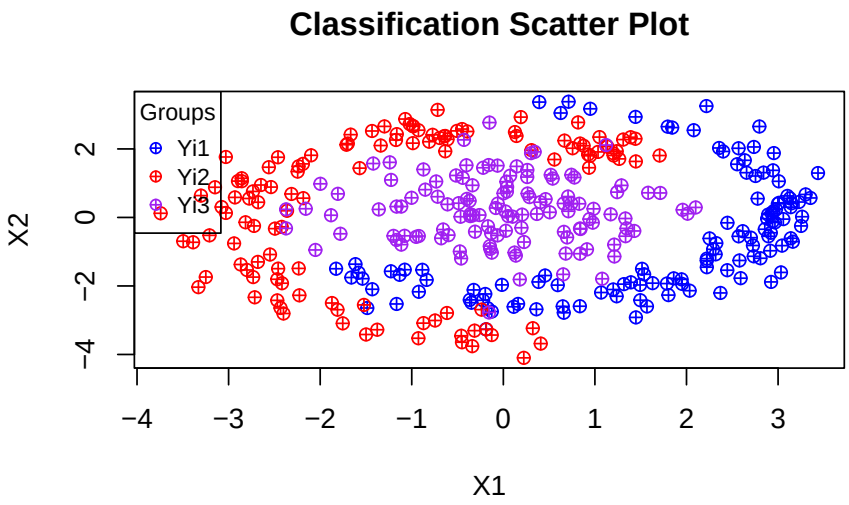


e) Interpretaion *comeback

- Linear Model has better OOS performance overall averaged across 1000 simulations and varying values of the noise parameter
- Non linear model has lower bias but higer variance and the ooposite is observed for variance of the model
- Given the estimated OOS Error, go with the linear model as it is more parsimonious

Q4 Feed Forward Neural Network

a) Data Set Plot



Observation: The data does not appear to be linearly separable (in the given 2D feature space) as the data does not seem to have linear boundries between categories/classes. A non-linear neural network seems well suited to capture the non-linear pattern/structure embedded in the data in for the given feature space.

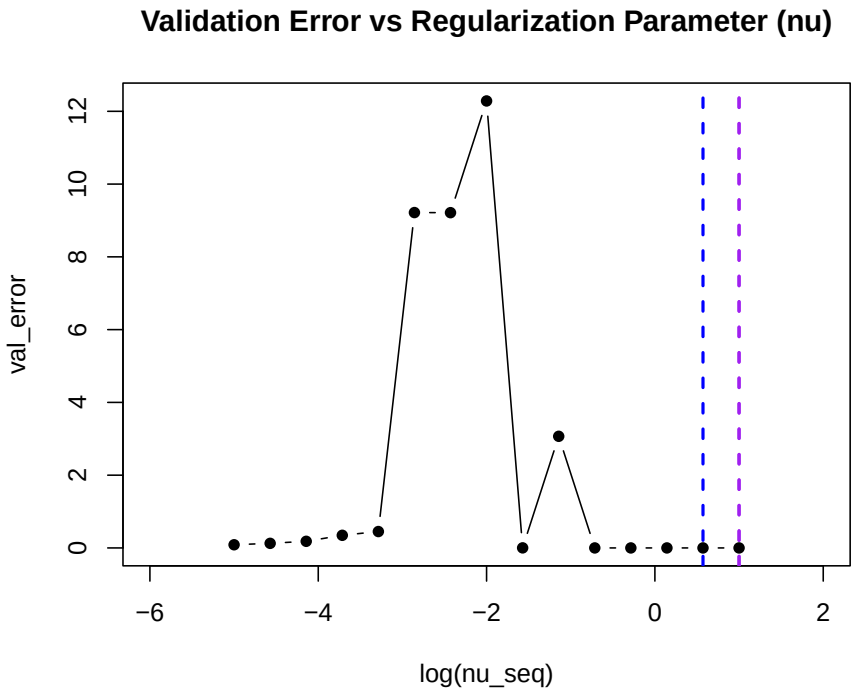
b)

c) Feed Forward Neural Network

Numerical Pitfalls: The computaion of log(yhat) in the objective function may cause the network to explode/ diverge given the undefined behavior of the log function at 0 (or values very close to 0), which can hamper the numerical stability of the network's computations.

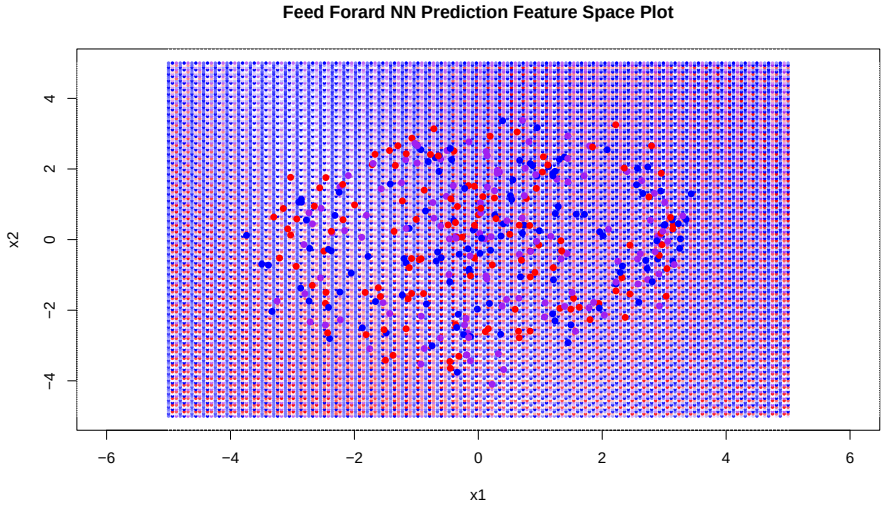
d) Validation Analysis

Plot the above



*comeback Lowest Validation error is obtained at nu=0.007. The closets nu value to the right of the obtained leads to an increase in the validation error, whilst the next best nu (in terms of validation error), produces and more complex model. Thus in this case the lowest nu stays!

d) Response Curve (?)


Feed Forard NN Prediction Feature Space Plot
x2
4
2
0
−2
−4
−6
−4
−2
0
2
4
6
x1